TECHNICAL NOTE • MHD-WAAM THESIS • PHASE 0

Chaotic Sensitivity in Melt-Pool CFD

Why cell-by-cell comparison cannot verify this class of simulation, and the noise-floor protocol that replaces it

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1. Summary

A verification exercise on a small laser melt-pool tutorial produced local temperature differences of up to 1008 K (36% of the field range) between a serial run and a 6-core parallel run of *identical* physics. A controlled perturbation experiment showed this is not a defect in the parallel implementation. The solution is **deterministically chaotic**: a perturbation at machine round-off level is amplified exponentially until it saturates at the scale of the field itself.

The consequence is methodological, and it constrains how every result in this thesis must be verified and reported. Point-by-point field comparison is not a valid test of correctness for this class of problem, and a local difference between two cases cannot by itself be attributed to a physical cause. This note records the evidence, states the constraint, and defines the protocol that makes claims about magnetic-field effects defensible.

> **THE CONSTRAINT IN ONE SENTENCE**
>
> Any difference between two simulations that is smaller than the numerical noise floor is not a result — and the noise floor must be measured, not assumed.

2. The observation

The test case was `Plate2D` from the laserbeamFoam tutorials: a 2D laser-melting problem on a 1 × 40 × 80 mesh (3,200 cells), run to t = 0.01 s with adaptive time stepping at maxCo = maxAlphaCo = 0.1. The physics comprises VOF interface tracking, melting with latent heat, enthalpy–porosity mushy-zone damping, Boussinesq buoyancy and temperature-dependent surface tension.

Three runs were compared on the temperature field:

- **Serial** — the reference.
- **Perturbed serial** — identical, except the `p_rgh` linear-solver tolerance was tightened from 1×10^{-7} to 1×10^{-8}. This changes nothing physical; it perturbs the solution only at round-off level.
- **Parallel** — 6 MPI ranks, scotch decomposition, then reconstructed.

Table 1 — Maximum absolute deviation from the serial reference

t (s)	PERTURBED SERIAL max deviation (K)	relative	PARALLEL (6 CORE) max deviation (K)	relative	ratio par / pert
0.001	1.395e-03	7.17e-07	8.026e-03	4.13e-06	5.8
0.003	6.743e-01	3.04e-04	5.575e+01	2.51e-02	83
0.005	1.524e+00	7.10e-04	1.215e+02	5.66e-02	80
0.010	8.924e+02	3.54e-01	1.008e+03	4.00e-01	1.1

Relative values are the maximum absolute deviation, divided by the range of the reference field (300 K to 2824 K). Both runs completed normally to t = 0.01 s.

Global quantities, by contrast, agree closely. Mean temperature was 679.04 K (serial) against 682.05 K (parallel), a difference of 0.44%, and the minimum was exactly 300.00 K in both — every cell the physics had not yet reached was bitwise identical. Total energy is conserved and essentially equal; only its local distribution differs.

3. Diagnosis: chaos, not a defect

Three features of Table 1 rule out an implementation error.

3.1 The parallel run begins at round-off

At t = 0.001 s the parallel deviation is 4.13×10^{-6} relative — round-off magnitude. A genuine parallel defect, such as a boundary condition mishandling processor faces or an incorrect global reduction, is wrong from the *first time step* and remains wrong at roughly constant magnitude. It does not begin at round-off and grow.

3.2 A serial perturbation reaches the same endpoint

This is the decisive evidence. By t = 0.01 s the perturbed *serial* run — which never invoked MPI — deviates by 3.54×10^{-1} relative, against 4.00×10^{-1} for the parallel run. Changing one linear-solver tolerance produces essentially the same divergence as decomposing across six cores. Parallel execution is therefore behaving as one perturbation among many, not as a source of error.

3.3 Growth is exponential, then saturates

The perturbed serial deviation grows from 1.4×10^{-3} K to 8.9×10^{2} K — nearly six orders of magnitude — monotonically. This is Lyapunov amplification. At t = 0.01 s both runs converge on a similar value because the temperature field is bounded (300–3000 K), so deviations cannot exceed the scale of the field. Saturation explains why the ratio in the final column collapses from 80 to 1.1.

The larger mid-range ratio has a straightforward explanation. Domain decomposition reorders floating-point summation in every gradient, divergence and global reduction at every time step, whereas a tolerance change perturbs only the pressure solve. Parallel injects a broader continuous perturbation; both then amplify at the same exponential rate.

4. Physical origin

This is not a numerical artefact to be eliminated. A melt pool driven by Marangoni convection with a deformable free surface is a convectively unstable system. The thermocapillary force depends on the surface temperature gradient, which depends on the flow the force itself drives — a positive feedback that amplifies infinitesimal differences. Adding droplet impingement, as the WAAM model will, introduces a further strongly nonlinear forcing.

The governing equations remain deterministic. Identical inputs on identical hardware with identical decomposition reproduce identical output. What is lost is *insensitivity*: the solution does not depend continuously on perturbations at round-off scale, so two numerically equivalent computations diverge locally while remaining statistically and globally equivalent.

WHAT REMAINS RELIABLE

Conserved and integral quantities. Total energy, mass, mean temperature, and time-averaged pool geometry are reproducible to within a fraction of a percent. It is only the instantaneous local field that is not.

5. The methodological constraint

Three practices are invalidated for this class of simulation.

INVALID	WHY	USE INSTEAD
Verifying a solver by comparing field values cell by cell against a reference	Two runs of identical physics differ by 1000 K locally. The comparison carries no information about correctness.	Integral and geometric measures: deposit height and width, penetration depth, total heat input, droplet radius and detachment frequency.
Attributing a local difference between two cases to a physical cause	The same difference arises between two runs of the *same* case. Causation cannot be inferred without a noise reference.	Establish the noise floor first; report only effects that exceed it (Section 6).
Reporting an instantaneous snapshot as a characteristic result	A velocity field at one instant is not reproducible under any perturbation.	Time-average over the quasi-steady window, and state the window.

THE QUESTION THIS ANTICIPATES

“How do you know that pool asymmetry is caused by the magnetic field, and not by numerical noise?”

Without a measured noise floor there is no answer to this, and every magnetic result in the thesis is open to the same objection. Section 6 is the answer.

6. The noise-floor protocol

For each model configuration, run the baseline case twice: once normally, and once with a perturbation at round-off level — a linear-solver tolerance changed by a factor of ten is sufficient, and changes no

physics. Extract every quantity the thesis will report from both runs. The spread between them is the noise floor for that quantity.

Interpretation is then unambiguous:

- An effect **several times larger** than the noise floor is a physical result and may be reported as such.
- An effect **comparable to** the noise floor is not a result. It must be reported as below the resolution of the method.
- The noise floor differs per quantity. Integral measures such as deposit width will have a low floor; local peak velocity will have a high one. Quote it per quantity, not once.

The cost is one additional run per model. In return, every comparison in the results chapter acquires an explicit resolution limit, in the same way that an experimental measurement is quoted with its uncertainty. This should be presented as a strength of the methodology, not a caveat buried in an appendix.

7. Consequences for the case matrix

RULE	REASON
Add a `-noise` twin of the baseline case for each model (D, P, T)	Supplies the noise floor. One extra run each.
Fix `n_procs` and `decomp_method` across every case that will be compared	Noise cannot be removed, but it need not be added to. Changing core count between cases introduces a perturbation that is indistinguishable from a physical effect.
Hold schemes, tolerances and mesh identical across a comparison set	Same reasoning. Vary one thing at a time: the magnetic field.
Record the solver commit hash and decomposition in `case_meta.json`	Reproducibility requires the full numerical configuration, not only the physical parameters.
Build a deliberately **non-chaotic** regression case — pure conduction, or Marangoni at low Ma	Serial and parallel must agree to ~1e-12 there. This is the only genuine test of parallel correctness, and it is what a chaotic case cannot provide. Run it after every change to the coded sources.

8. How to state this in the thesis

Place this in the methodology chapter, before results, as a subsection on verification. The argument in four sentences:

SUGGESTED FRAMING

The melt-pool solution is deterministically chaotic, and perturbations at machine precision are amplified to the scale of the field within the simulated interval. Verification against point values is therefore invalid, and all reported quantities are integral or time-averaged measures. For each model configuration a noise floor was established by repeating the baseline case with a round-off-level perturbation. Only differences exceeding that floor are attributed to the applied magnetic field.

Stating the limitation explicitly, with the supporting experiment, is stronger than omitting it. An examiner who raises chaotic sensitivity unprompted will find it already addressed and quantified.

9. Checklist

	ACTION	PHASE
1	Add `n_procs` and `decomp_method` columns to caseMatrix.csv; emit a generated run script per case	1
2	Add `-noise` twin rows for the baseline of each model	1
3	Build the non-chaotic parallel regression case into the waamMhdFoam test suite	2
4	Define the reported quantity list, and confirm every entry is integral or time-averaged	3
5	Measure and tabulate the noise floor per quantity before interpreting any magnetic case	5
6	State the constraint and the protocol in the methodology chapter	7

Reproducing the experiment

Case directories `P2D-ser`, `P2D-ser2` and `P2D-par` under `~/thesis/cases/00-tutorials/`. The comparison tool is `compare_runs.py` in the same directory:

```
python3 compare_runs.py T 0.001 0.003 0.005 0.01 \
    -- P2D-ser P2D-ser2 P2D-par
```

The perturbation is applied with `foamDictionary system/fvSolution -entry solvers/p_rgh/tolerance -set 1e-08`.